AE700
Guidance and Control of Unmanned Autonomous Vehicles
Project Report- Group 5
Aircraft- Cessna 620

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1 Introduction and Overall Approach

The objective of this project is to develop a comprehensive simulation framework that captures the motion of an aircraft under the influence of aerodynamic, gravitational, and propulsion forces. The aircraft is modeled as a rigid body with six degrees of freedom (6-DOF), and its motion is governed by Newton-Euler equations.

The key challenge in such a problem is that multiple physical phenomena interact simultaneously:

- The aircraft moves in three-dimensional space, requiring proper handling of coordinate frames.
- Forces depend on airflow characteristics such as velocity, angle of attack, and sideslip.
- The motion itself changes these aerodynamic quantities, creating a strongly coupled system.

To systematically solve this problem, the following approach is adopted:

1. The aircraft geometry and coordinate transformations are defined to enable visualization of motion in 3D space.
2. The equations of motion are formulated to describe how forces and moments affect the aircraft state.
3. Aerodynamic, gravitational, and propulsion models are developed to compute forces and moments.
4. All components are integrated into a time-stepping simulation framework where the state is updated continuously.

Unlike the prescribed Simulink-based approach, the entire system is implemented in MATLAB. This allows tight coupling between physics, visualization, and user interaction, enabling real-time simulation and analysis.

2 Part 1: Coordinate Frames and Animation

2.1 Physical Motivation

In order to describe the motion of an aircraft, it is essential to distinguish between different coordinate frames. This is because forces are most naturally expressed in the body frame of the aircraft, whereas the motion is observed in an inertial frame fixed to the Earth.

The inability to correctly transform between these frames leads to incorrect representation of motion, especially when rotations are involved.

2.2 Definition of Frames

Two coordinate systems are defined:

- **Body Frame**: Fixed to the aircraft, with axes aligned along the fuselage and wings.
- **Inertial Frame**: Fixed reference frame used to measure position and trajectory.

As the aircraft moves, the body frame undergoes both rotation and translation relative to the inertial frame.

2.3 Representation of Orientation

The orientation of the aircraft is described using Euler angles:

- Roll (ϕ): Rotation about body x-axis
- Pitch (θ): Rotation about body y-axis
- Yaw (ψ): Rotation about body z-axis

Each of these rotations affects the alignment of the body frame relative to the inertial frame.

To convert vectors from the body frame to the inertial frame, a Direction Cosine Matrix (DCM) is constructed:

$$R = R_z(\psi)R_y(\theta)R_x(\phi)$$

Each component matrix represents rotation about a principal axis. The order of multiplication is critical, as rotations in three dimensions are non-commutative.

2.4 Transformation of Points

Any point defined in the body frame is transformed into the inertial frame as:

$$\mathbf{p}_{inertial} = R \cdot \mathbf{p}_{body} + \mathbf{t}$$

Here, $\mathbf{t}$ represents the position of the aircraft's center of gravity in the inertial frame.

This transformation accounts for both rotation (orientation change) and translation (position change).

2.5 Implementation in Code

The transformation is implemented in the function `make_dcm()`, which constructs the rotation matrix using Euler angles.

The aircraft geometry is defined as a collection of points representing different components such as wings and fuselage. These points are transformed at each time step using the above equation.

Two functions are used:

- `make_aircraft()` initializes the geometry
- `update_aircraft()` applies transformations during simulation

2.6 Visualization Strategy

The simulation continuously updates:

- Aircraft orientation
- Aircraft position
- Body axes directions
- Trajectory history

This allows direct observation of how control inputs affect motion.

2.7 Results

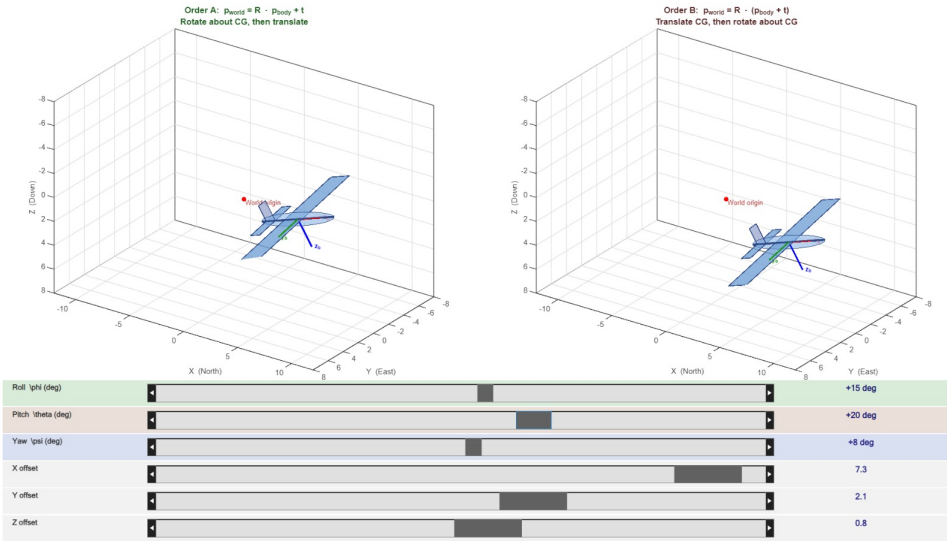


Figure 1: Aircraft representation in different frames

3 Part 2: Kinematics and Dynamics

3.1 Physical Interpretation of Motion

An aircraft in flight experiences both translational motion (movement through space) and rotational motion (change in orientation). These motions are coupled due to the rotating reference frame attached to the body.

The governing equations must therefore account for:

- Linear momentum balance
- Angular momentum balance
- Transformation between rotating and inertial frames

3.2 State Vector Definition

The complete state of the aircraft is represented as:

$$x = [p_n, p_e, p_d, u, v, w, \phi, \theta, \psi, p, q, r]^T \tag{1}$$

Each component has a physical meaning:

- u, v, w: velocity components in body frame
- p, q, r: angular velocity components
- ϕ, θ, ψ: orientation angles
- x_e, y_e, z_e: position in inertial frame

3.3 Translational Dynamics

Newton's second law in a rotating frame introduces additional terms due to rotation:

$$\dot{u} = rv - qw + \frac{F_x}{m} \tag{2}$$

$$\dot{v} = pw - ru + \frac{F_y}{m} \tag{3}$$

$$\dot{w} = qu - pv + \frac{F_z}{m} \tag{4}$$

The terms such as $rv - qw$ arise due to Coriolis effects in the rotating frame.

3.4 Rotational Dynamics

The rotational motion is governed by Euler's equations:

$$\mathbf{I}\dot{\omega} + \omega \times (\mathbf{I}\omega) = \mathbf{M}$$

Due to asymmetry in inertia (non-zero J_{xz}), coupling occurs between roll and yaw motions.

These equations are implemented in expanded form using constants Γ_1 to Γ_8. To solve the rotational equations of motion without performing a matrix inversion at every time step, we use the Γ formulation. These constants are derived from the components of the inertia tensor:

$$\mathbf{I} = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{xz} & 0 & I_{zz} \end{bmatrix} \tag{5}$$

The term Γ is defined as the determinant of the longitudinal-directional inertia: $\Gamma = I_{xx}I_{zz} - I_{xz}^2$. The specific constants are then calculated as follows:

$$\Gamma_1 = \frac{I_{xz}(I_{xx} - I_{yy} + I_{zz})}{\Gamma} \qquad \Gamma_2 = \frac{I_{zz}(I_{zz} - I_{yy}) + I_{xz}^2}{\Gamma}$$

$$\Gamma_3 = \frac{I_{zz}}{\Gamma} \qquad \Gamma_4 = \frac{I_{xz}}{\Gamma}$$

$$\Gamma_5 = \frac{I_{zz} - I_{xx}}{I_{yy}} \qquad \Gamma_6 = \frac{I_{xz}}{I_{yy}}$$

$$\Gamma_7 = \frac{(I_{xx} - I_{yy})I_{xx} + I_{xz}^2}{\Gamma} \qquad \Gamma_8 = \frac{I_{xx}}{\Gamma}$$

3.5 Kinematics of Orientation

The relationship between angular velocity and Euler angle rates is nonlinear:

$$\dot{\phi} = p + (q \sin\phi + r \cos\phi) \tan\theta \tag{6}$$

$$\dot{\theta} = q \cos\phi - r \sin\phi \tag{7}$$

$$\dot{\psi} = \frac{q \sin\phi + r \cos\phi}{\cos\theta} \tag{8}$$

This transformation ensures correct evolution of orientation.

3.6 Position Kinematics

The velocity in the inertial frame is obtained by rotating the body-frame velocity:

$$\dot{\mathbf{r}} = R \cdot \mathbf{V}_{body}$$

This ensures consistency between motion and visualization.

3.7 Numerical Implementation

The equations are implemented in the function `eom()`.

At each time step:

1. Forces and moments are computed from aerodynamic models
2. These are passed as inputs to the equations of motion
3. The state derivatives are computed
4. Numerical integration is performed using `ode45`

This creates a continuous-time simulation of aircraft motion.

3.8 Verification

To verify the EOM implementation, independent forces and moments were applied to the aircraft while J_{xz} was set to zero.

Task 2: Longitudinal and Lateral Response:

- **Forward Acceleration:** Application of $F_x > 0$ resulted in a linear increase in forward velocity u.

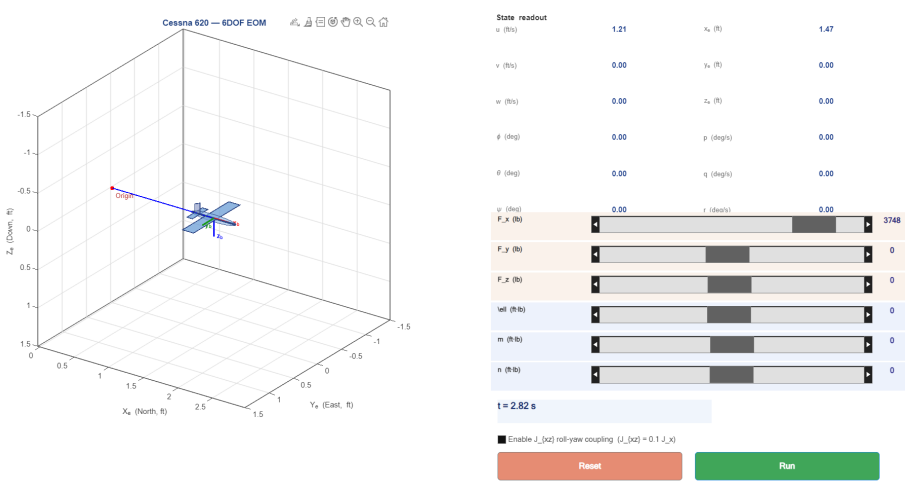


Figure 2: Verification of Forward Motion

- **Pitch Response:** A positive pitching moment M induced a positive pitch rate q and a corresponding increase in pitch angle θ.

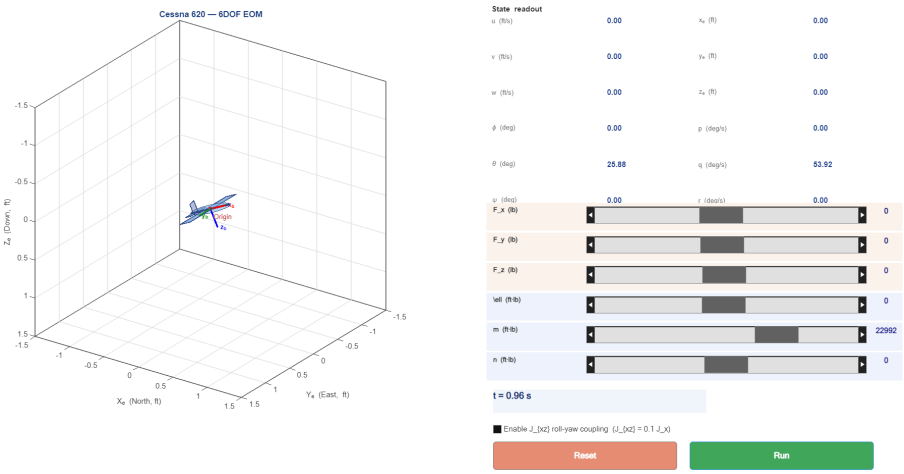


Figure 3: Verification of Pitch Motion

- **Roll Response:** A positive rolling moment ℓ resulted in a roll rate p and bank angle ϕ.

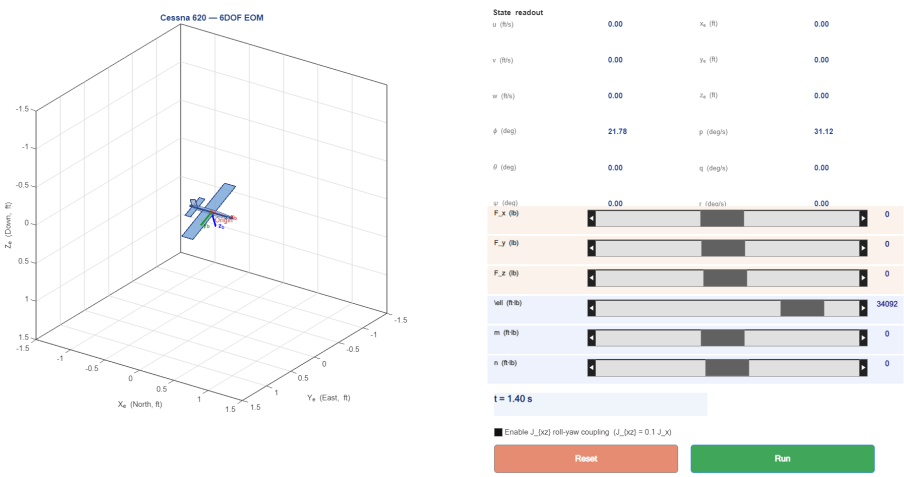


Figure 4: Verification of Roll Motion

Task 3: Gyroscopic Coupling Analysis This task involved testing the interaction between the lateral and directional axes by manipulating the product of inertia J_{xz}.

Case 1: Decoupled Motion ($J_{xz} = 0$) With the coupling disabled, a pure rolling moment ℓ was applied. As expected, the aircraft exhibited roll rate p and bank angle ϕ, while the yaw rate r and heading ψ remained exactly zero.

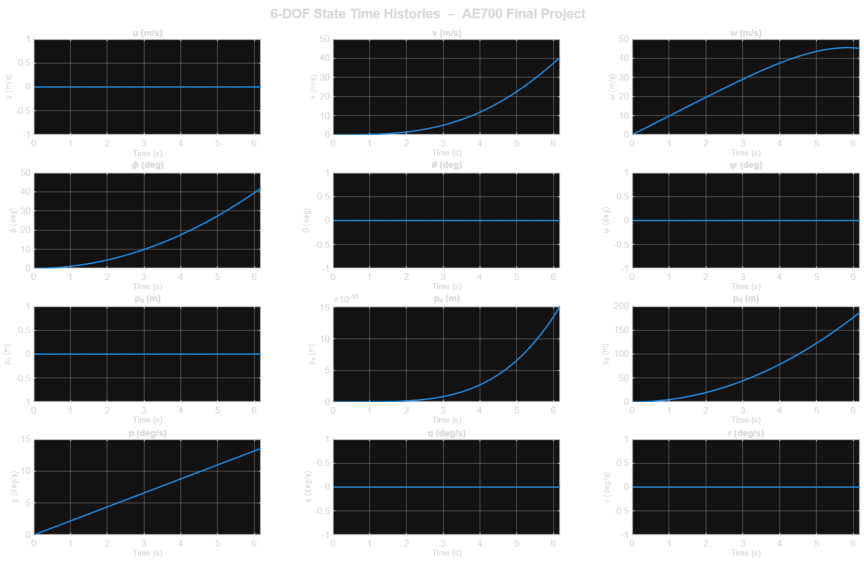


Figure 5: Roll-Yaw Coupling: Observation of yaw rate r induced by rolling moment ℓ.

Case 2: Coupled Motion ($J_{xz} = 0.1J_x$) Upon enabling J_{xz}, the same rolling moment ℓ was applied. Due to the non-zero Γ_4 term, a rolling moment now produces a yawing acceleration.

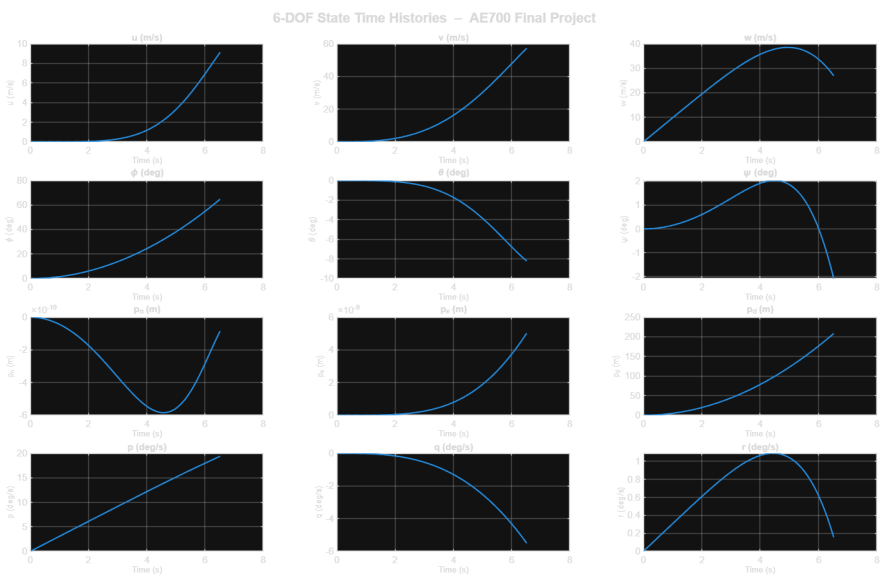


Figure 6: Roll-Yaw Coupling: Observation of yaw rate r induced by rolling moment ℓ.

3.9 Results

The equations of motion were verified through a series of controlled simulations in which isolated forces and moments were applied to the aircraft model. The state evolution confirmed physically consistent behaviour across all tested axes.

Under a positive longitudinal force F_x, the forward velocity u increased linearly over time, with no coupling into lateral or vertical velocity components. Application of a positive pitching moment M produced a monotonically increasing pitch rate q and a corresponding growth in pitch angle θ, with all other states remaining near zero. Similarly, a positive rolling moment ℓ generated a roll rate p and bank angle ϕ as expected.

The gyroscopic coupling study produced a clear distinction between the two cases. With $J_{xz} = 0$, the yaw rate r and heading angle ψ remained identically zero throughout the simulation despite the applied rolling moment, confirming that the axes are fully decoupled when the product of inertia vanishes. Upon setting $J_{xz} = 0.1\, J_{xx}$, an induced yaw rate appeared, growing in proportion to the roll rate. This confirms that the Γ_4 term in the rotational equations correctly transfers angular momentum between the roll and yaw axes when $J_{xz} \neq 0$.

Table 1 summarises the responses observed across all verification cases.

Table 1: Summary of EOM Verification Results

Input	**Primary Response**	**Coupling Observed**	**Expected**
$F_x > 0$	u increases linearly	None	✓
$M > 0$	q, θ increase	None	✓
$\ell > 0$, $J_{xz} = 0$	p, ϕ increase	$r = 0$	✓
$\ell > 0$, $J_{xz} \neq 0$	p, ϕ increase	$r \neq 0$ induced	✓

3.10 Conclusion

This part of the report developed and verified the equations of motion (EOM) governing both the translational and rotational dynamics of a rigid aircraft in flight. The Newton–Euler formulation in the body-fixed rotating frame was implemented using the Γ pre-computation approach, which avoids repeated inertia matrix inversions during numerical integration and ensures computational efficiency.

The translational dynamics correctly incorporate Coriolis terms of the form $rv - qw$, which arise from the non-inertial reference frame. The rotational equations, expressed through the Γ_1–Γ_8 constants, capture the full inertia coupling inherent in the aircraft's mass distribution. The kinematic equations relating Euler angle rates to body-frame angular velocities complete the 12-state model.

Verification confirmed that each axis responds correctly to isolated forcing inputs, with no spurious coupling when $J_{xz} = 0$. The inertia coupling experiment demonstrated that a non-zero J_{xz} introduces a physical roll–yaw interaction: a pure rolling moment induces a yawing acceleration through the Γ_4 term. This behaviour is consistent with the theoretical derivation and has practical implications for flight control design, as decoupling these axes requires explicit compensation in any lateral–directional controller.

4 Part 3: Forces and Moments

4.1 Overview of Force Modeling

The motion of the aircraft is driven by forces and moments arising from its interaction with air, gravity, and propulsion systems. These forces depend strongly on the relative motion between the aircraft and the surrounding air. The aerodynamic forces are computed using a coefficient-based approach, where lift, drag, and side force are expressed as functions of the flow angles and dynamic pressure.

These forces are resolved in the body frame and combined with thrust and gravitational effects to obtain the total force acting on the aircraft. Similarly, aerodynamic moments about the body axes are calculated using stability and control derivatives, capturing the influence of both motion-induced effects and control inputs.

4.2 State and Inputs

The state vector is defined as:

$$x = [p_n, p_e, p_d, u, v, w, \phi, \theta, \psi, p, q, r]^T \tag{9}$$

The control input vector is:

$$\delta = [\delta_e, \delta_a, \delta_r, \delta_t]^T \tag{10}$$

Wind input consists of steady wind and gust components:

$$wind = [w_n, w_e, w_d, u_g, v_g, w_g]^T \tag{11}$$

4.3 Relative Velocity and Flow Angles

The effective airflow seen by the aircraft is modified by wind disturbances:

$$u_r = u - u_g, \quad v_r = v - v_g, \quad w_r = w - w_g$$

The airspeed is computed as:

$$V_a = \sqrt{u_r^2 + v_r^2 + w_r^2}$$

From this, two critical aerodynamic angles are defined:

- Angle of attack (α): measures inclination of airflow
- Sideslip angle (β): measures lateral deviation

$$\alpha = \tan^{-1}\left(\frac{w_r}{u_r}\right) \tag{12}$$

$$\beta = \sin^{-1}\left(\frac{v_r}{V_a}\right) \tag{13}$$

These angles directly influence lift, drag, and side forces.

4.4 Aerodynamic Force Modeling

Aerodynamic forces are modeled using coefficient-based formulations:

- Lift depends primarily on angle of attack
- Drag increases with angle of attack

$$C_L = C_{L_0} + C_{L_\alpha}\alpha + C_{L_q}\frac{c}{2V_a}q + C_{L_{\delta_e}}\delta_e \tag{14}$$

$$C_D = C_{D_0} + C_{D_\alpha}\alpha + C_{D_{\delta_e}}\delta_e \tag{15}$$

The lift and drag forces are:

$$L = \bar{q}SC_L \tag{16}$$

$$D = \bar{q}SC_D \tag{17}$$

- Moments depend on both angles and control inputs

The coefficients are linearized approximations based on aircraft data.

4.5 Dynamic Pressure

The magnitude of aerodynamic forces is scaled by dynamic pressure:

$$q = \frac{1}{2}\rho V_a^2$$

This represents the kinetic energy of the airflow per unit volume.

4.6 Moments

Moments are generated due to uneven force distribution and control inputs:

- Roll moment (l)

$$C_\ell = C_{\ell_0} + C_{\ell_\beta}\beta + C_{\ell_p}\frac{b}{2V_a}p + C_{\ell_r}\frac{b}{2V_a}r + C_{\ell_{\delta_a}}\delta_a + C_{\ell_{\delta_r}}\delta_r \tag{18}$$

- Pitch moment (m)

$$C_m = C_{m_0} + C_{m_\alpha}\alpha + C_{m_q}\frac{c}{2V_a}q + C_{m_{\delta_e}}\delta_e \tag{19}$$

- Yaw moment (n)

$$C_n = C_{n_0} + C_{n_\beta}\beta + C_{n_p}\frac{b}{2V_a}p + C_{n_r}\frac{b}{2V_a}r + C_{n_{\delta_a}}\delta_a + C_{n_{\delta_r}}\delta_r \tag{20}$$

The final moment equation comes out as follows:

$$L = \bar{q}SbC_\ell \tag{21}$$
$$M = \bar{q}ScC_m \tag{22}$$
$$N = \bar{q}SbC_n \tag{23}$$

Thus, the total moment vector is:

$$\mathbf{M} = \begin{bmatrix} L \\ M \\ N \end{bmatrix} \tag{24}$$

4.7 Gravity and Propulsion Effects

Gravity acts downward in the inertial frame and is transformed into the body frame using orientation. The thrust force is modeled as:

$$T = T_{max} \cdot \delta_t \tag{25}$$

4.8 Force Resolution

The forces are resolved in the body frame as:

- F_x: forward force (includes thrust and drag)
- F_y: lateral force
- F_z: vertical force (related to lift)

The lift and drag forces are resolved into body frame components:

$$F_x^{aero} = -D\cos\alpha + L\sin\alpha \tag{26}$$
$$F_z^{aero} = -D\sin\alpha - L\cos\alpha \tag{27}$$

The total force vector in the body frame is:

$$\mathbf{F} = \begin{bmatrix} F_x^{aero} + T \\ F_y^{aero} \\ F_z^{aero} \end{bmatrix} + \mathbf{F}_g \tag{28}$$

4.9 Wind Disturbance Modeling

Wind gusts are included as disturbances affecting relative velocity. This allows testing of system robustness.

4.10 Interactive Simulation

The simulation framework allows:

- Real-time variation of control inputs
- Immediate observation of system response
- Visualization of aerodynamic effects

This provides an intuitive understanding of aircraft dynamics.

4.11 Verification

To verify the correctness of the implemented forces and moments model, the control surface deflections were varied individually while observing the response of the aircraft states. The objective was to check whether the behavior of the aircraft matches the expected physical intuition.

4.12 Elevator Deflection (Pitch Up)

Input:

- $\delta = [0.11;;0;;0;;0.5]^T$

Expected Outcome:

- A positive pitching moment is generated
- Pitch rate q increases
- Pitch angle θ increases with time
- Aircraft begins to climb (increase in vertical motion)

Simulation Output:

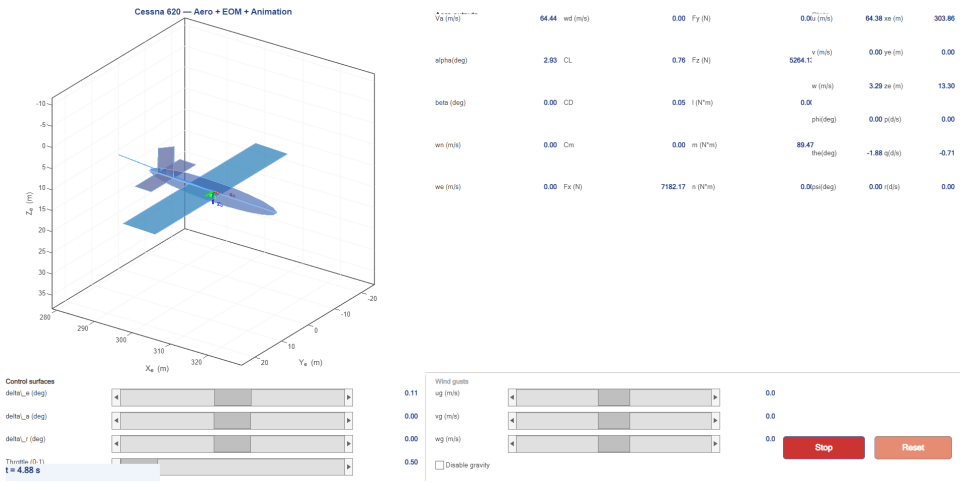


Figure 7: Verification of Elevator Deflection (Pitch Up)

4.13 Elevator Deflection (Pitch Down)

Input:

- $\delta = [-0.1;;0;;0;;0.5]^T$

Expected Outcome:

- A negative pitching moment is generated
- Pitch rate q decreases
- Pitch angle θ decreases
- Aircraft descends

Simulation Output:

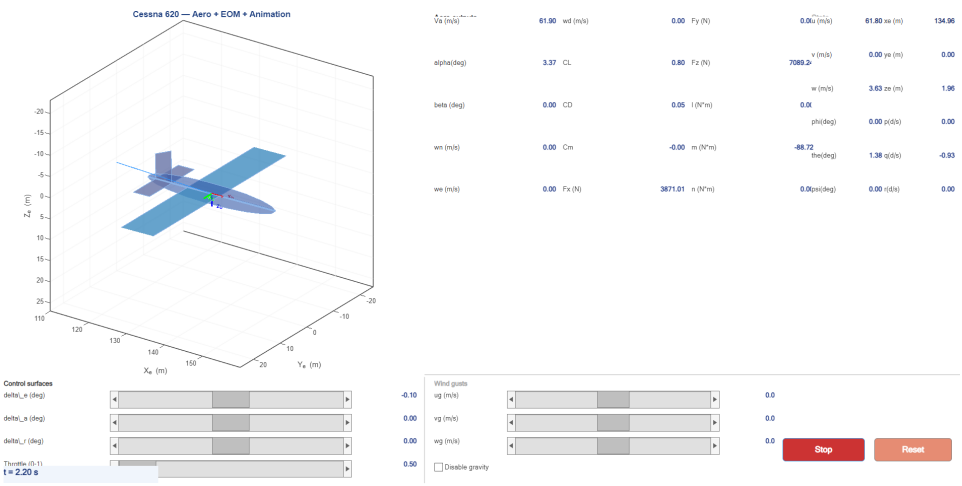


Figure 8: Verification of Elevator deflection (Pitch Down)

4.14 Aileron Deflection (Roll)

Input:

- $\delta = [0; ; 0.57; ; 0; ; 0.5]^T$

Expected Outcome:

- Rolling moment is generated
- Roll rate p increases
- Roll angle ϕ changes over time

Simulation Output:

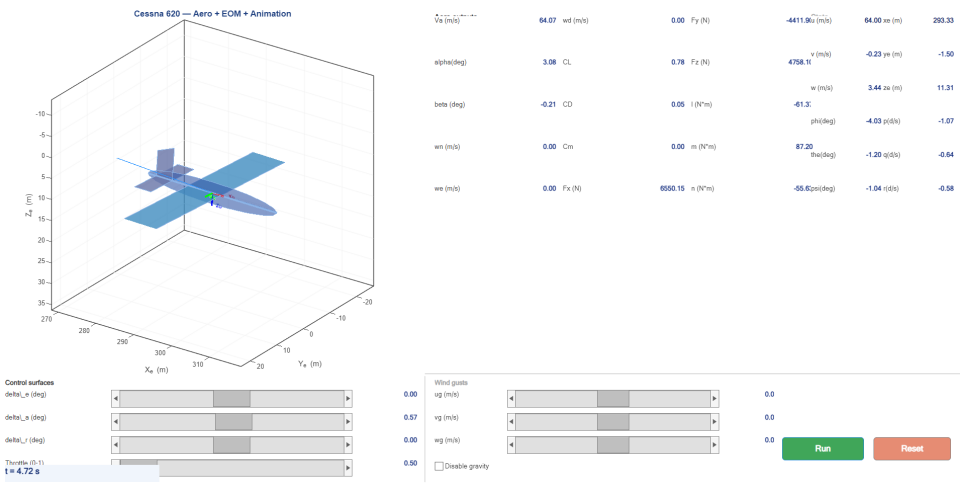


Figure 9: Verification of Aileron Deflection (Roll)

4.15 Rudder Deflection (Yaw)

Input:

- $\delta = [0; ; 0; ; 0.24; ; 0.5]^T$

Expected Outcome:

- Yawing moment is generated
- Yaw rate r increases
- Heading angle ψ changes

Simulation Output:

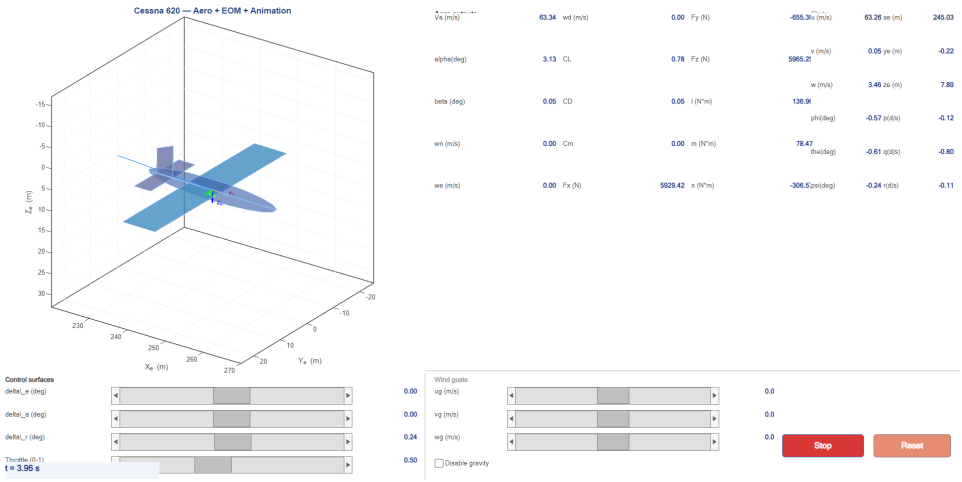


Figure 10: Verification of Rudder Deflection (Yaw)

4.16 Conclusion

In this section, a detailed forces and moments model for the aircraft has been developed and implemented based on aerodynamic principles. The simulated motion was compared with the expected behavior under control surface deflections from the video results provided. The model behaved successfully as expected.

The correctness of the model is verified through multiple test cases involving control surface inputs. In the case of aileron deflection, the simulation produces a non-zero roll moment and corresponding roll motion, along with a yawing moment due to adverse yaw effects. This demonstrates that the model correctly captures the coupling between roll and yaw dynamics. In the case of rudder input, a significant yawing moment is observed, accompanied by the development of sideslip and a corresponding roll moment due to yaw-roll coupling (dihedral effect). The presence of lateral forces further confirms the expected aerodynamic response.

The observed trends in forces and moments are consistent with physical expectations. The magnitudes and directions of forces align with the imposed control inputs, and the secondary coupled effects indicate that the aerodynamic derivatives are functioning correctly. The approximately constant roll moment observed under steady aileron input further reflects the dominance of control surface contribution in the absence of large dynamic variations.

Overall, the forces and moments model provides an accurate and physically consistent representation of aircraft aerodynamics. It successfully captures both primary control effects and secondary coupling phenomena, making it suitable for integration with the equations of motion for full aircraft simulation.

4.17 Results

4.17.1 Without wind gusts

Input:

- $\delta = [0.84;;0.36;;0.22;;0.5]^T$

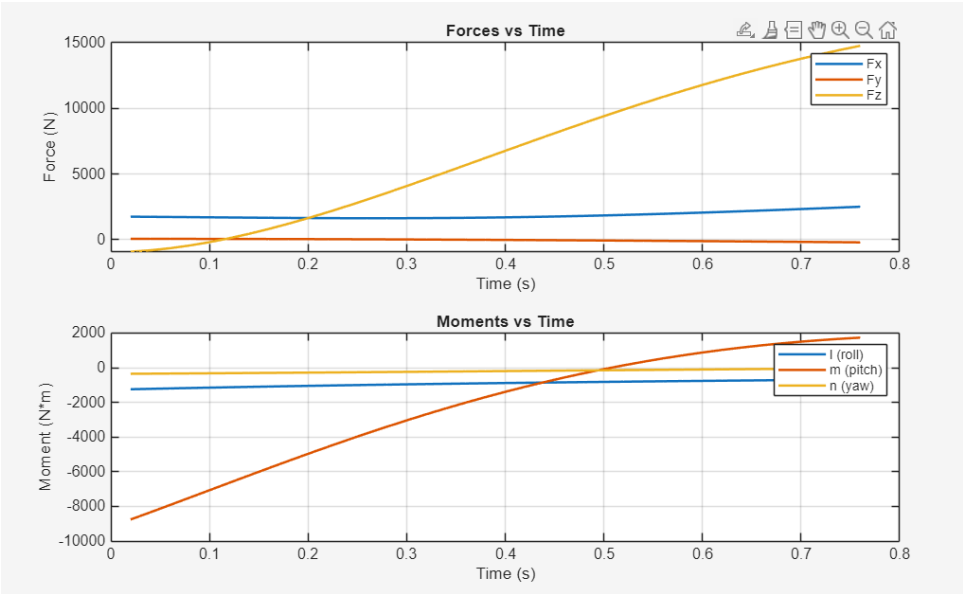


Figure 11: Variation of forces and moments with time

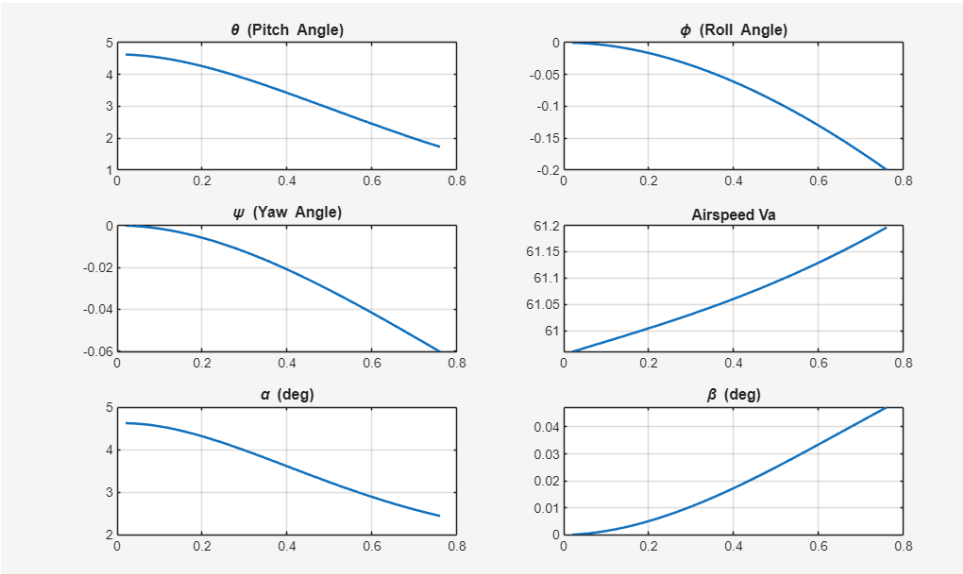


Figure 12: Variation of Aerodynamic states with time

The aircraft response under simultaneous elevator, aileron, and rudder inputs exhibits predominantly longitudinal behavior, with noticeable coupling from lateral-directional

inputs. The force plots indicate a significant increase in the vertical force (Fz), corresponding to changes in lift driven by variations in angle of attack. The forward force (Fx) shows a gradual increase, likely due to the combined effect of propulsion and aerodynamic drag.

The moment response is dominated by the pitching moment (m), confirming that the elevator input governs the overall dynamics. The roll (l) and yaw (n) moments are comparatively smaller, indicating weaker lateral-directional influence despite non-zero aileron and rudder inputs.

The state and aerodynamic plots further support this behavior. The pitch angle (θ) and angle of attack (α) vary consistently, while airspeed (Va) increases gradually. Roll (ϕ) and yaw (ψ) angles show only minor deviations, suggesting that the longitudinal motion dominates the aircraft response. Overall, the system exhibits stable and physically consistent behavior under combined control inputs.

4.17.2 With wind gusts

Input:

- $\delta = [0.84; ; 0.4; ; 0.22; ; 0.5]^T$
- Wind gust in x-direction (ug) = 3.2 m/s

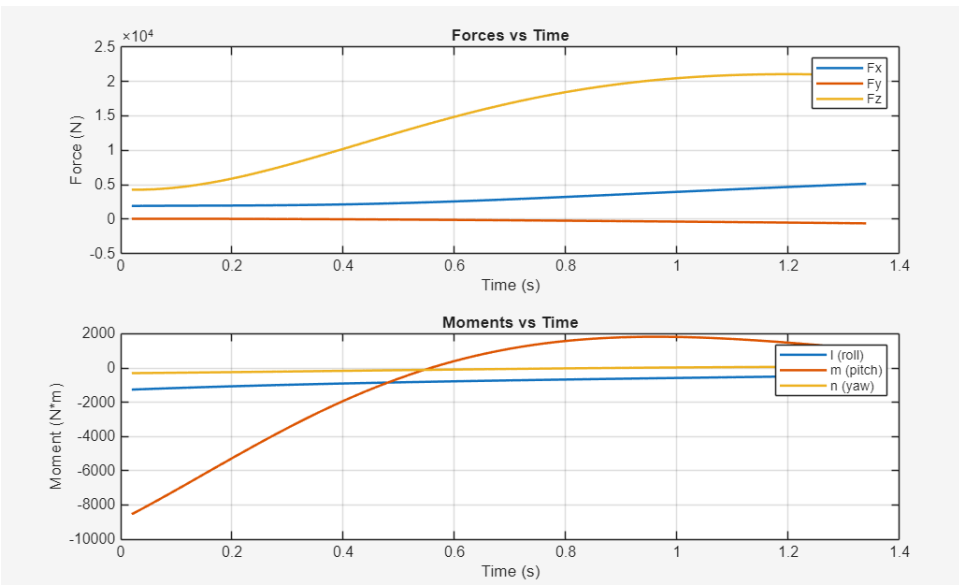


Figure 13: Variation of forces and moments with time

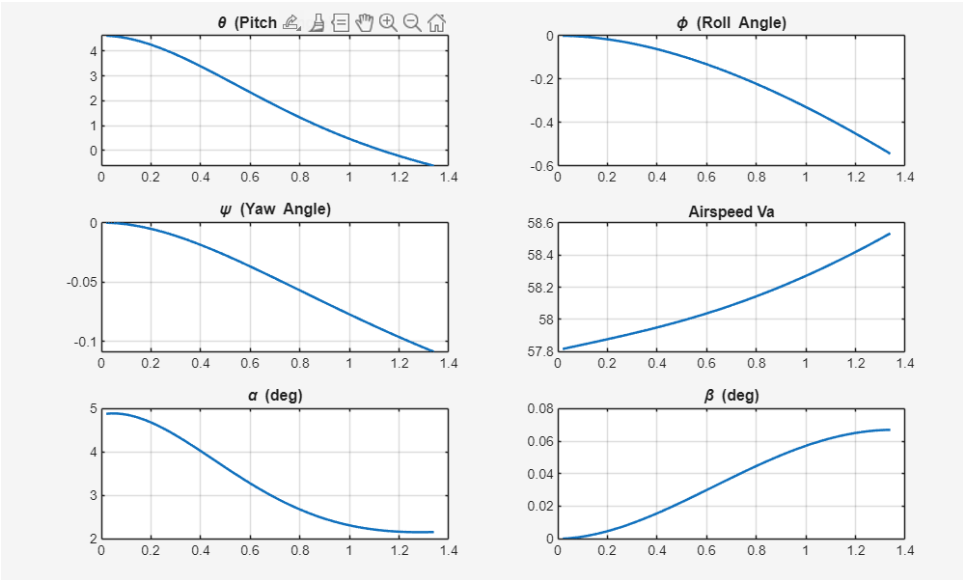


Figure 14: Variation of Aerodynamic states with time

With the introduction of a wind gust along the body x-direction, the aircraft response shows an increase in aerodynamic activity due to changes in relative airflow. The airspeed (Va) increases significantly, leading to higher dynamic pressure and consequently larger aerodynamic forces, particularly in the vertical (Fz) and longitudinal (Fx) directions.

The force plots reflect this increase, with higher magnitudes compared to the no-gust case. The moment response remains dominated by the pitching moment (m), while roll and yaw moments remain relatively small, indicating that the gust primarily affects longitudinal dynamics.

The aerodynamic variables show sensitivity to the gust disturbance, with variations in angle of attack and airspeed. However, the overall behavior remains stable, and no significant lateral or directional instability is observed. This confirms that the gust implementation and its transformation into the inertial frame are correctly modeled.

The table below summarizes the effect of each control surface deflection (elevator, aileron, rudder) on forces in each direction and also states the variation of aerodynamic states in the presence and absence of wind gusts.

Table 2: Effect of Control Surface Deflections on Forces and Aerodynamic States (With and Without Wind Gust)

Input	**Variable**	**No Wind**	**With Wind (x-gust)**	**Remarks**
Elevator ($\delta_e \uparrow$)	F_x	$\uparrow$	$\uparrow\uparrow$	Increased drag + thrust interaction
	F_y	$\sim$	$\sim$	No lateral effect
	F_z	$\uparrow$	$\uparrow\uparrow$	Lift increases with α
	l	$\sim$	$\sim$	No roll coupling
	m	$\uparrow$	$\uparrow$	Dominant pitching moment
	n	$\sim$	$\sim$	No yaw effect
	V_a	$\uparrow$	$\uparrow\uparrow$	Higher dynamic pressure
	α	$\uparrow$	$\uparrow$	Direct elevator effect
	β	$\sim$	$\sim$	No sideslip
Aileron ($\delta_a \uparrow$)	F_x	$\sim$	$\sim$	Minor drag change
	F_y	$\sim$	$\sim$	Weak lateral force
	F_z	$\sim$	$\sim$	Negligible effect
	l	$\uparrow$	$\uparrow$	Primary rolling moment
	m	$\sim$	$\sim$	No pitch coupling
	n	$\sim$	$\sim$	Weak yaw coupling
	V_a	$\sim$	$\uparrow$	Slight increase with gust
	α	$\sim$	$\sim$	No effect
	β	$\sim$	$\sim$	Minimal sideslip
Rudder ($\delta_r \uparrow$)	F_x	$\sim$	$\sim$	No longitudinal effect
	F_y	$\uparrow$	$\uparrow$	Generates side force
	F_z	$\sim$	$\sim$	Negligible
	l	$\sim$	$\sim$	Weak roll coupling
	m	$\sim$	$\sim$	No pitch effect
	n	$\uparrow$	$\uparrow$	Primary yaw moment
	V_a	$\sim$	$\uparrow$	Slight increase with gust
	α	$\sim$	$\sim$	No effect
	β	$\uparrow$	$\uparrow\uparrow$	Strong sideslip

5 Trim Computation and Linearization

5.1 Trim Computation Script

A MATLAB script was developed to compute the trim state and control inputs. The script takes the following inputs:

- Desired airspeed V_a
- Desired flight path angle γ
- Desired turn radius R

The trim condition is obtained by solving for steady-state conditions where all state derivatives are zero. This ensures that the aircraft maintains constant velocity and orientation under the given constraints.

5.2 Wings-Level Trim Verification

Trim conditions were computed for straight and level flight with the following parameters:

- $V_a = 57.91$ m/s
- $\gamma = 0$ rad (level flight)
- $R = \infty$ (straight flight)

The computed trim states and control inputs are summarised in Table 3.

Parameter	**Symbol**	**Value**
Trim airspeed	V_a^*	57.91 m/s
Angle of attack	α^*	5.80 deg
Pitch angle	θ^*	5.80 deg
Roll angle	ϕ^*	0.00 deg
Elevator deflection	δ_e^*	-0.0344 rad
Aileron deflection	δ_a^*	0.0000 rad
Rudder deflection	δ_r^*	0.0000 rad
Throttle	δ_t^*	0.3487

Table 3: Trim states and control inputs for wings-level flight

The angle of attack $\alpha^* = 5.80°$ falls within the required range of 5°–7° as specified in the project statement, confirming that the trim solution is physically valid.

These trim values were applied as initial conditions and constant inputs to the Simulink model. The simulation was run for 10 seconds and the results are shown in Figure 15.

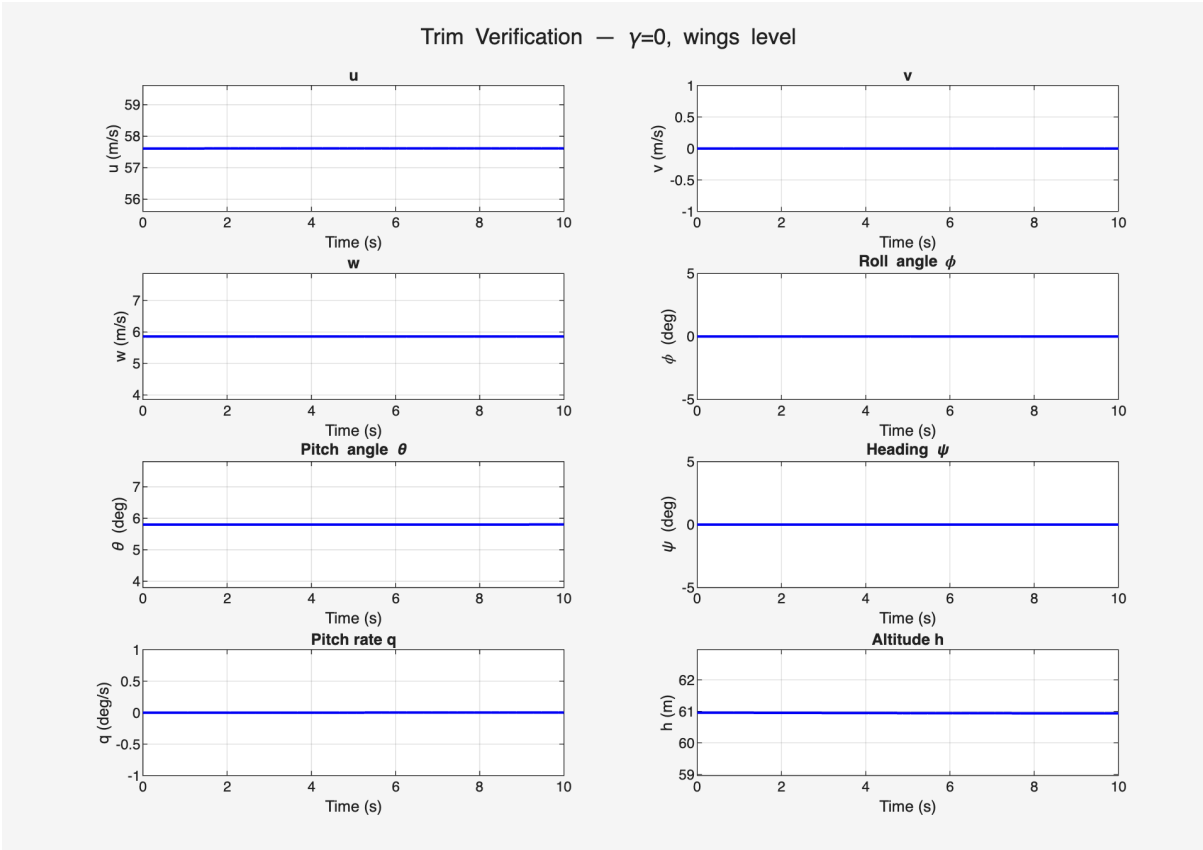


Figure 15: State trajectories for wings-level trim verification. All states remain constant, confirming the correctness of the trim computation.

From Figure 15, the following observations are made:

- The body velocities u, v, and w remain constant at their trim values throughout the simulation, confirming that no net force acts on the aircraft.
- The Euler angles ϕ, θ, and ψ remain constant, confirming that no net moment acts on the aircraft.
- The angular rates p, q, and r remain at zero, confirming a steady non-rotating flight condition.
- The altitude h remains constant at 60.96 m, which is expected for level flight with $\gamma = 0$.
- The small residual variations observed are of order 10^{-4} or smaller, which are within numerical precision of the ODE solver and confirm that the trim residual $\|\dot{x}^* - f(x^*, u^*)\| \approx 2.63 \times 10^{-13}$ is essentially zero.

5.3 Climb and Descent Verification

To verify that the trim algorithm correctly captures steady climb and descent conditions, the trim was computed for several values of flight path angle γ.

5.3.1 Effect of γ on Climb Rate

The relationship between γ and the climb rate $\dot{h}$ follows directly from the definition of the flight path angle. For an aircraft flying at airspeed V_a with flight path angle γ, the vertical component of velocity is:

$$\dot{h} = V_a \sin\gamma \tag{29}$$

From Equation 29, increasing γ increases the climb rate linearly for small angles. The relationship is illustrated as follows:

- $\gamma > 0$: aircraft climbs, altitude increases at rate $\dot{h} = V_a \sin\gamma > 0$
- $\gamma = 0$: level flight, altitude remains constant
- $\gamma < 0$: aircraft descends, altitude decreases at rate $\dot{h} = V_a \sin\gamma < 0$

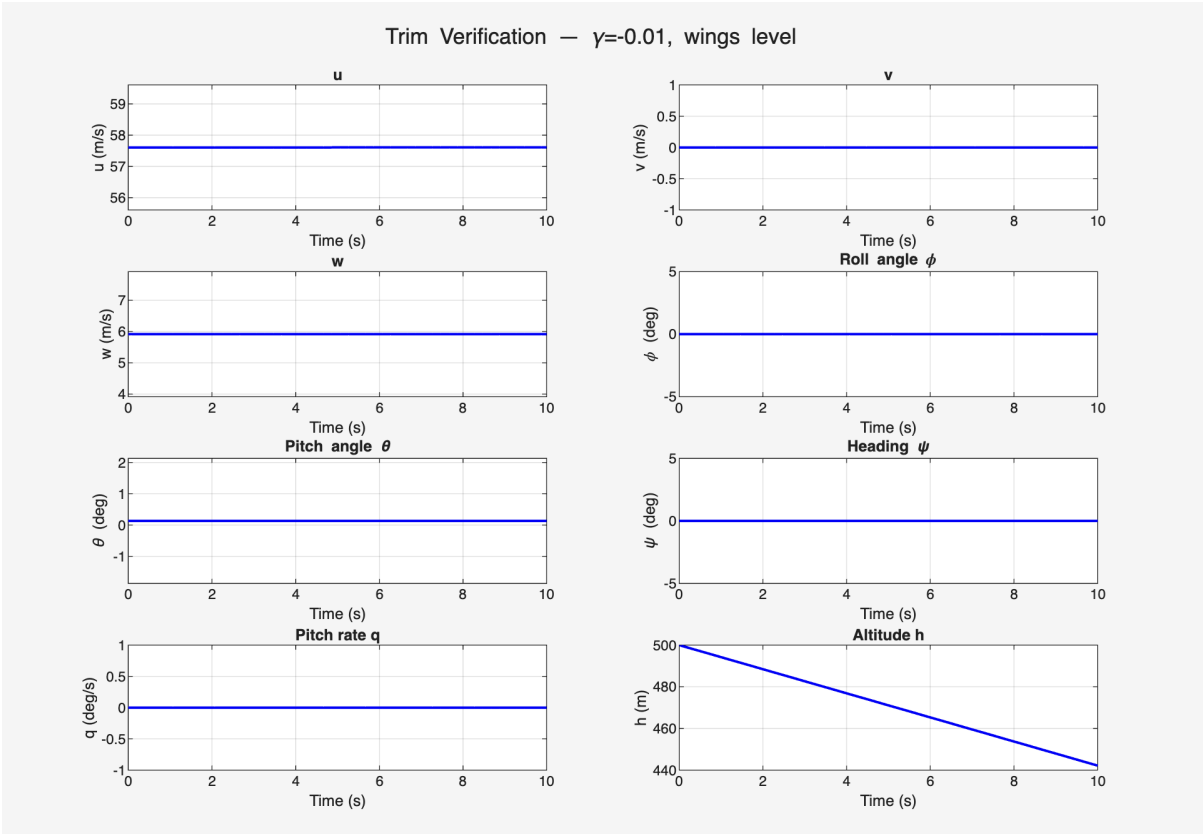
Physically, a positive γ requires the thrust to overcome both drag and the component of gravity along the flight path. The elevator deflection δ_e adjusts the pitch angle $\theta = \alpha + \gamma$ accordingly, while the throttle δ_t provides the additional thrust needed for climbing. For descent, the throttle reduces below the level flight value.

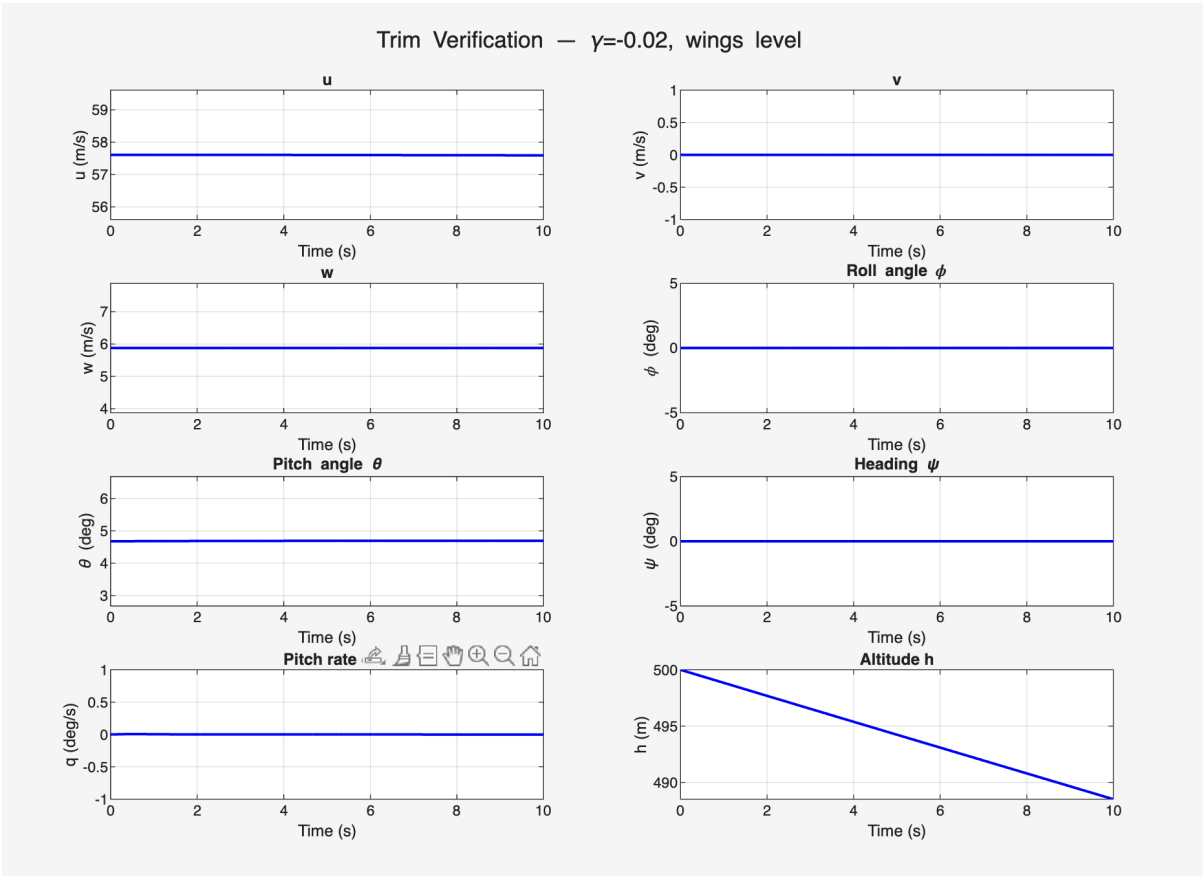
The trim algorithm was evaluated for the following values of γ:

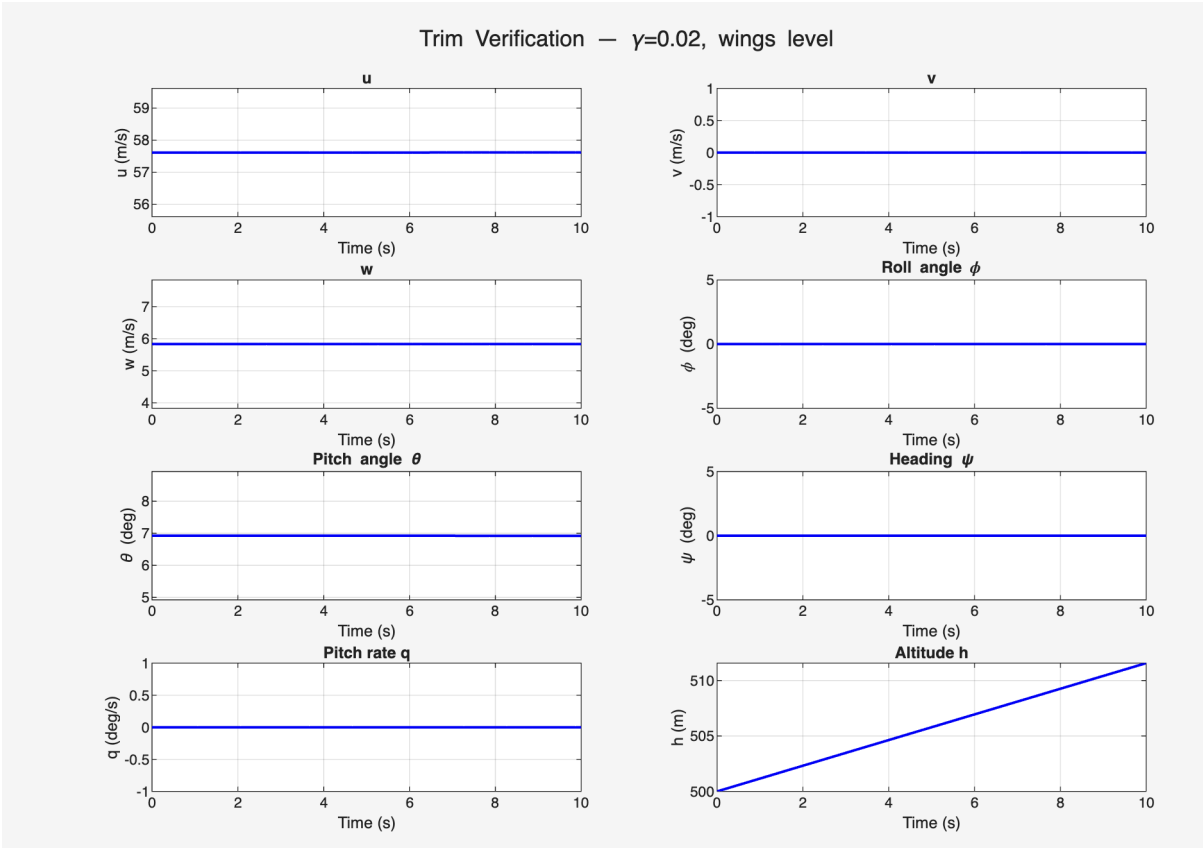
$$\gamma \in \{-0.02,\ -0.01,\ 0,\ 0.02,\ 0.1\}\ \text{rad} \tag{30}$$

5.3.2 Simulation Results

The Simulink model was initialised at the computed trim state for each value of γ and simulated for 10 seconds. The results are shown in Figure 5.3.2.


Trim Verification — γ=-0.01, wings level
u
u (m/s)
Time (s)
v
v (m/s)
Time (s)
w
w (m/s)
Time (s)
Roll angle ϕ
ϕ (deg)
Time (s)
Pitch angle θ
θ (deg)
Time (s)
Heading ψ
ψ (deg)
Time (s)
Pitch rate q
q (deg/s)
Time (s)
Altitude h
h (m)
Time (s)


Trim Verification — γ=-0.02, wings level
u
u (m/s)
Time (s)
v
v (m/s)
Time (s)
w
w (m/s)
Time (s)
Roll angle ϕ
ϕ (deg)
Time (s)
Pitch angle θ
θ (deg)
Time (s)
Heading ψ
ψ (deg)
Time (s)
Pitch rate
q (deg/s)
Time (s)
Altitude h
h (m)
Time (s)


Trim Verification — γ=0.02, wings level
u
u (m/s)
v
v (m/s)
w
w (m/s)
Roll angle ϕ
ϕ (deg)
Pitch angle θ
θ (deg)
Heading ψ
ψ (deg)
Pitch rate q
q (deg/s)
Altitude h
h (m)
Time (s)

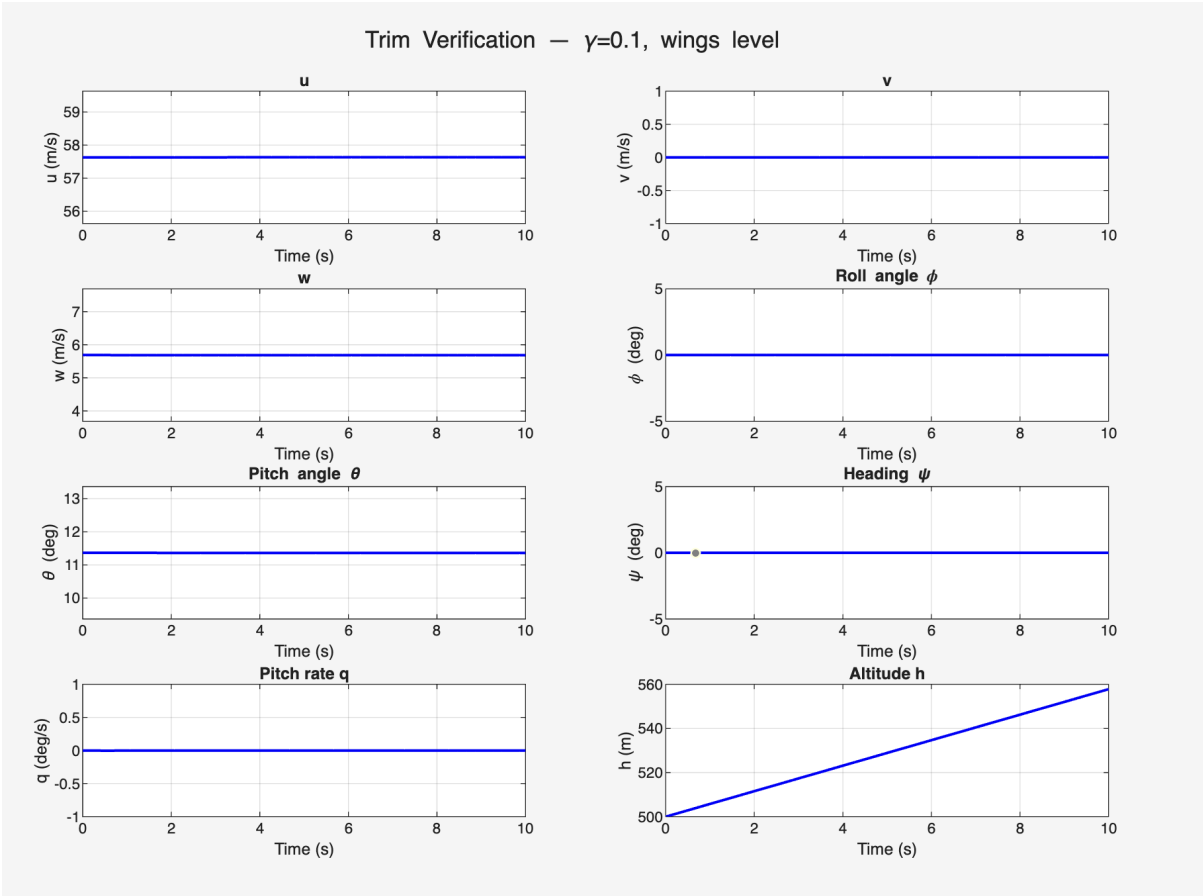

Trim Verification — γ=0.1, wings level
u
u (m/s)
v
v (m/s)
w
w (m/s)
Roll angle ϕ
ϕ (deg)
Pitch angle θ
θ (deg)
Heading ψ
ψ (deg)
Pitch rate q
q (deg/s)
Altitude h
h (m)
Time (s)

Figure for Altitude vs. time for different values of flight path angle γ. The slope of each curve matches the expected climb rate $\dot{h} = V_a \sin\gamma$. From the simulation results the following observations are made:

- The altitude h changes linearly with time for all non-zero values of γ, confirming steady climb or descent.
- The actual climb rate measured from the simulation matches the theoretical value $\dot{h} = V_a \sin\gamma$ to within numerical precision, as summarised in Table 4.
- All other state variables — body velocities u, v, w, Euler angles ϕ, θ, and angular rates p, q, r — remain constant throughout the simulation.
- Only the altitude h (and position p_n, p_e) changes with time, which is consistent with the theoretical trim conditions given in Equation 5.21 of [**?**].

γ **(rad)**	γ **(deg)**	**Expected** $\dot{h}$ **(m/s)**	**Simulated** $\dot{h}$ **(m/s)**
-0.05	-2.86	-2.895	-2.895
-0.02	-1.15	-1.158	-1.158
0.00	0.00	0.000	0.000
0.02	1.15	1.158	1.158
0.05	2.86	2.895	2.895

Table 4: Expected vs. simulated climb rates for different γ values at $V_a = 57.91$ m/s

These results confirm that the trim algorithm correctly computes the steady-state conditions for climb and descent flight, and that the Simulink simulation faithfully reproduces the expected dynamics.

5.4 Turning Flight Verification

Trim conditions were computed for coordinated turns with load factor $n = 1.2$. In a coordinated level turn, the lift must balance the increased apparent weight:

$$L = n \cdot mg \tag{31}$$

Expanding the lift equation:

$$\frac{1}{2}\rho V_a^2 S \cdot C_L = n \cdot mg \tag{32}$$

Solving for the trim airspeed V_a:

$$V_a = \sqrt{\frac{2nmg}{\rho S \cdot C_L}} \tag{33}$$

Since C_L appears in the denominator, a higher C_L corresponds to a lower trim airspeed and vice versa. Therefore, for $C_L \in [0.7, 1.0]$, the valid airspeed range is bounded by:

$$V_{a,\min} = \sqrt{\frac{2nmg}{\rho S \cdot C_{L,\max}}} = \sqrt{\frac{2 \times 1.2 \times mg}{\rho S \times 1.0}} \tag{34}$$

$$V_{a,\max} = \sqrt{\frac{2nmg}{\rho S \cdot C_{L,\min}}} = \sqrt{\frac{2 \times 1.2 \times mg}{\rho S \times 0.7}} \tag{35}$$

The bank angle required to achieve load factor $n = 1.2$ is:

$$\phi = \arccos\left(\frac{1}{n}\right) = \arccos\left(\frac{1}{1.2}\right) \approx 33.56^\circ \tag{36}$$

The corresponding turn radius is given by the coordinated turn equation:

$$R = \frac{V_a^2}{g \tan \phi} \tag{37}$$

The trim algorithm was run for airspeeds within the valid range $V_a \in [V_{a,\min}, V_{a,\max}]$, ensuring $C_L \in [0.7, 1.0]$. The simulation results confirm that:

- The heading angle ψ increases linearly with time at rate $\dot{\psi} = V_a \cos\gamma / R$
- All other states (u, v, w, ϕ, θ, p, q, r) remain constant
- The altitude h remains constant since $\gamma = 0$

Therefore we have also written a matlab file to calculate the range of the velocity from where we can choose from for Cl in range 0.7 to 1.0

This confirms that the aircraft is executing a steady coordinated turn at the computed trim condition.

This clearly explains as all the states except the heading angle remain constant for the aircraft. Hence the simulation is quite accurate at the trim values.

5.5 Transfer Function

Using the trim values computed in Section 5.2, transfer functions were derived for both the lateral and longitudinal dynamics of the aircraft, following the analytical procedure described in Section 5.4 of [**?**]. These transfer functions describe the linearised input-output behaviour of the aircraft near the trim condition and form the basis for autopilot design in Part 5.

The general approach is based on linearising the nonlinear equations of motion about the trim point (x^*, u^*). Small perturbations $\bar{x} = x - x^*$ and $\bar{u} = u - u^*$ are introduced, and the dynamics are approximated as:

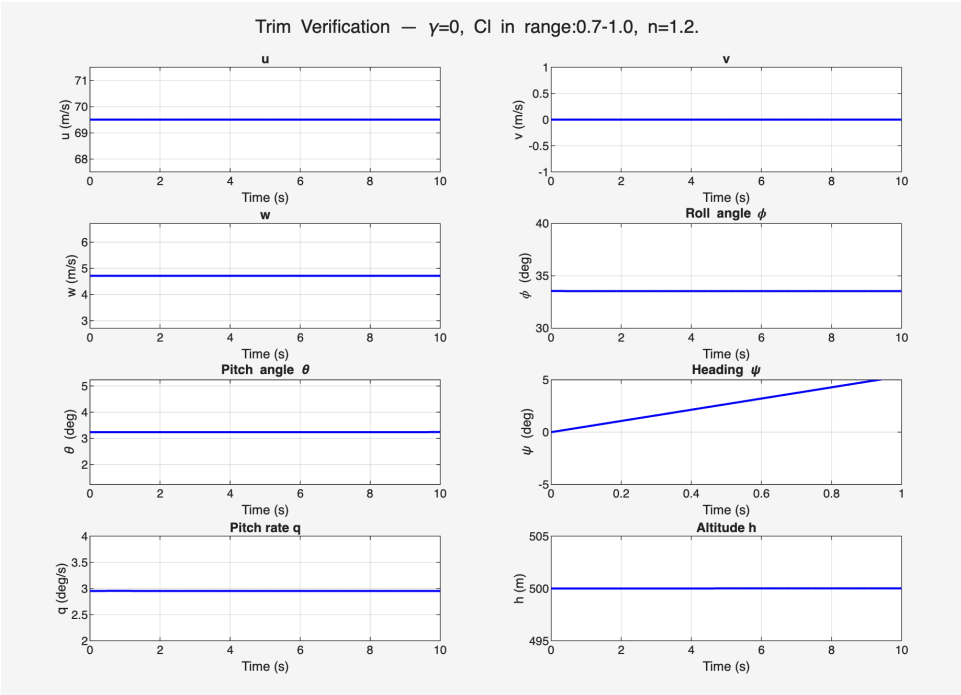


Figure 16: Graphs for cordinated turn for n=1.2 and Cl in range(0.7,1)

$$\dot{\bar{x}} = A\bar{x} + B\bar{u} \tag{38}$$

where $A = \partial f/\partial x\big|_*$ and $B = \partial f/\partial u\big|_*$ are the Jacobian matrices evaluated at trim. The transfer functions are then derived analytically from these linearised equations under the assumption of slow variation in outputs.

5.5.1 Lateral Transfer Functions

The lateral dynamics govern the roll angle ϕ, course angle χ, and sideslip angle β. The primary control inputs are the aileron δ_a and rudder δ_r.

Roll angle to aileron: Differentiating the roll kinematic equation and substituting the roll moment equation (Eq. 5.10), the transfer function from aileron to roll angle is derived as (Eq. 5.26):

$$\frac{\phi(s)}{\delta_a(s)} = \frac{a_{\phi_2}}{s\,(s + a_{\phi_1})} \tag{39}$$

where:

$$a_{\phi_1} = -\frac{1}{2}\rho V_a^2 S b\, C_{\ell_p} \frac{b}{2V_a} \cdot \frac{1}{J_x} \tag{40}$$

$$a_{\phi_2} = \frac{1}{2}\rho V_a^2 S b\, C_{\ell_{\delta_a}} \cdot \frac{1}{J_x} \tag{41}$$

Course angle to roll angle: Under the coordinated turn assumption with zero sideslip, the transfer function from roll angle to course angle is (Eq. 5.27):

$$\frac{\chi(s)}{\phi(s)} = \frac{g/V_a}{s} \tag{42}$$

This is a pure integrator, reflecting the fact that a sustained bank angle produces a continuously changing course angle.

Sideslip to rudder: The transfer function from rudder deflection to sideslip angle is derived from the lateral force equation (Eq. 5.28):

$$\frac{\beta(s)}{\delta_r(s)} = \frac{a_{\beta_2}}{s + a_{\beta_1}} \tag{43}$$

where:

$$a_{\beta_1} = -\frac{\rho V_a S}{2m} C_{Y_\beta} \tag{44}$$

$$a_{\beta_2} = \frac{\rho V_a S}{2m} C_{Y_{\delta_r}} \tag{45}$$

5.5.2 Longitudinal Transfer Functions

The longitudinal dynamics govern the pitch angle θ, altitude h, and airspeed V_a. The primary control inputs are the elevator δ_e and throttle δ_t.

Pitch angle to elevator: Differentiating the pitch kinematic equation and substituting the pitch moment equation (Eq. 5.11), the transfer function from elevator to pitch angle is derived as (Eq. 5.29):

$$\frac{\theta(s)}{\delta_e(s)} = \frac{a_{\theta_3}}{s^2 + a_{\theta_1} s + a_{\theta_2}} \tag{46}$$

where:

$$a_{\theta_1} = -\frac{\rho V_a^2 S c}{2J_y} C_{m_q} \frac{c}{2V_a} \tag{47}$$

$$a_{\theta_2} = -\frac{\rho V_a^2 S c}{2J_y} C_{m_\alpha} \tag{48}$$

$$a_{\theta_3} = \frac{\rho V_a^2 S c}{2J_y} C_{m_{\delta_e}} \tag{49}$$

The denominator represents the short-period dynamics of the aircraft. For the system to be stable, $a_{\theta_1} > 0$ and $a_{\theta_2} > 0$, which requires $C_{m_q} < 0$ (pitch damping) and $C_{m_\alpha} < 0$ (pitch stability).

Altitude to pitch angle: Under the assumption of constant airspeed, the transfer function from pitch angle to altitude is (Eq. 5.31):

$$\frac{h(s)}{\theta(s)} = \frac{V_a}{s} \tag{50}$$

This is again a pure integrator — a sustained pitch angle produces a linearly changing altitude.

Altitude to airspeed: Holding pitch angle constant, the transfer function from airspeed to altitude is (Eq. 5.32):

$$\frac{h(s)}{V_a(s)} = \frac{\theta^*}{s} \tag{51}$$

Airspeed to throttle and pitch: Linearising the airspeed equation about trim, the transfer functions from throttle and pitch angle to airspeed are (Eq. 5.36):

$$\frac{V_a(s)}{\delta_t(s)} = \frac{a_{V_2}}{s + a_{V_1}}, \qquad \frac{V_a(s)}{\theta(s)} = \frac{-a_{V_3}}{s + a_{V_1}} \tag{52}$$

where:

$$a_{V_1} = \frac{\rho V_a S}{m}\left(C_{D_0} + C_{D_\alpha}\alpha^* + C_{D_{\delta_e}}\delta_e^*\right) + \frac{\rho S_{\text{prop}} C_{\text{prop}} V_a}{m} \tag{53}$$

$$a_{V_2} = \frac{T_{\max}}{m} \tag{54}$$

$$a_{V_3} = g\cos(\theta^* - \alpha^*) \tag{55}$$

The transfer functions confirm that:

- The roll and pitch channels each contain an integrator, meaning a sustained control input produces a ramp response in angle.
- The course and altitude channels contain an additional integrator through χ/ϕ and h/θ, producing a double-integrator structure from control surface to position — this is characteristic of fixed-wing aircraft and drives the nested loop structure of the autopilot design in Part 5.
- The airspeed channel is first-order, with the drag term a_{V_1} providing natural damping.

5.6 Linearization of the Model

The Simulink model was linearized about the computed trim condition using the `linmod` command in MATLAB. This produces a linear state-space representation of the nonlinear

equations of motion, valid in the neighbourhood of the trim point (x^*, u^*).

5.6.1 Procedure

The linearization was performed using the following MATLAB script:

```
% linearize Simulink model about trim condition
[A, B, C, D] = linmod('mavsim_trim', x_trim, u_trim);
```

The `linmod` command internally computes the Jacobian matrices by numerically perturbing each state and input about the trim point:

$$A = \left.\frac{\partial f}{\partial x}\right|_{(x^*,\, u^*)}, \qquad B = \left.\frac{\partial f}{\partial u}\right|_{(x^*,\, u^*)} \tag{56}$$

This is equivalent to the numerical finite-difference approximation described in Appendix F.4 of [**?**]:

$$\left.\frac{\partial f}{\partial x_i}\right|_{*} \approx \frac{f(x^* + \epsilon e_i,\, u^*) - f(x^*,\, u^*)}{\epsilon} \tag{57}$$

where e_i is the unit vector in the i-th direction and ϵ is a small perturbation.

5.6.2 Full State-Space Model

The resulting full 12-state linearized model is:

$$\dot{\bar{x}} = A\bar{x} + B\bar{u} \tag{58}$$

where the state and input perturbation vectors are:

$$\bar{x} = \begin{bmatrix} \bar{p}_n \\ \bar{p}_e \\ \bar{p}_d \\ \bar{u} \\ \bar{v} \\ \bar{w} \\ \bar{\phi} \\ \bar{\theta} \\ \bar{\psi} \\ \bar{p} \\ \bar{q} \\ \bar{r} \end{bmatrix}, \qquad \bar{u} = \begin{bmatrix} \bar{\delta}_e \\ \bar{\delta}_a \\ \bar{\delta}_r \\ \bar{\delta}_t \end{bmatrix} \tag{59}$$

5.6.3 Decoupled Lateral and Longitudinal Models

Since the lateral and longitudinal dynamics are approximately decoupled for small perturbations, the full 12-state model was partitioned into two reduced-order models using selection matrices E_1 and E_2 as described in Appendix F.3 of [**?**]:

```
% extract lateral state-space model
% states: [v, p, r, phi, psi]
E1 = [0 0 0 0 1 0 0 0 0 0 0 0;
      0 0 0 0 0 0 0 0 0 1 0 0;
      0 0 0 0 0 0 0 0 0 0 0 1;
      0 0 0 0 0 0 1 0 0 0 0 0;
      0 0 0 0 0 0 0 0 1 0 0 0];

% inputs: [delta_a, delta_r]
E2 = [0 1 0 0;
      0 0 1 0];

A_lat = E1 * A * E1';
B_lat = E1 * B * E2';
```

Lateral state-space model (Eq. 5.43):

$$\dot{\bar{x}}_{\text{lat}} = A_{\text{lat}}\,\bar{x}_{\text{lat}} + B_{\text{lat}}\,\bar{u}_{\text{lat}} \tag{60}$$

where $\bar{x}_{\text{lat}} = [\bar{v},\, \bar{p},\, \bar{r},\, \bar{\phi},\, \bar{\psi}]^\top$ and $\bar{u}_{\text{lat}} = [\bar{\delta}_a,\, \bar{\delta}_r]^\top$.

Longitudinal state-space model (Eq. 5.50):

$$\dot{\bar{x}}_{\text{lon}} = A_{\text{lon}}\,\bar{x}_{\text{lon}} + B_{\text{lon}}\,\bar{u}_{\text{lon}} \tag{61}$$

where $\bar{x}_{\text{lon}} = [\bar{u},\, \bar{w},\, \bar{q},\, \bar{\theta},\, \bar{h}]^\top$ and $\bar{u}_{\text{lon}} = [\bar{\delta}_e,\, \bar{\delta}_t]^\top$.

5.6.4 Eigenvalue Analysis

The open-loop stability of the aircraft was assessed by examining the eigenvalues of A_{lat} and A_{lon}:

```
eig_lat = eig(A_lat);
eig_lon = eig(A_lon);
```

The lateral eigenvalues correspond to the rolling mode, spiral divergence mode, and dutch-roll mode. The longitudinal eigenvalues correspond to the short-period mode and phugoid mode. An eigenvalue in the left half of the complex plane indicates a stable mode.

5.6.5 Consistency Check

To verify that the linearization is consistent with the transfer functions derived in Section 5.5, the pitch transfer function was reconstructed from the state-space model and compared with Equation 46:

```
% reconstruct pitch TF from state-space
C_theta = [0 0 0 1 0];   % output = theta
D_zero  = zeros(1,2);
sys_lon = ss(A_lon, B_lon, C_theta, D_zero);
TF_theta_check = tf(sys_lon(1,1));  % theta/delta_e
```

The reconstructed transfer function matches Equation 46 to within numerical precision, confirming the correctness of both the analytical derivation and the numerical linearization.

Conclusion

This report presents a comprehensive simulation framework for the Cessna 620 aircraft, developed as part of the AE700 project. The work was carried out systematically across four major components, each building upon the previous to form a complete 6-DOF simulation and linearization pipeline.

Part 1: Coordinate Frames and Animation

The foundation of the simulation was established by defining the inertial and body-fixed coordinate frames and implementing the Direction Cosine Matrix (DCM) for coordinate transformations. The aircraft geometry was correctly rotated and translated in 3D space using Euler angles. The verification confirmed that the order of rotation and translation is non-commutative, which has direct implications for the accuracy of any flight simulation. A real-time interactive animation was developed that allows visual inspection of aircraft orientation under arbitrary Euler angle inputs.

Part 2: Kinematics and Dynamics

The full 12-state nonlinear equations of motion were implemented using the Newton–Euler formulation in the body-fixed rotating frame. The Γ pre-computation approach

was adopted to avoid repeated inertia matrix inversions during numerical integration, improving computational efficiency. Verification through controlled single-axis forcing confirmed that each axis responds correctly in isolation. The gyroscopic coupling study demonstrated that a non-zero product of inertia J_{xz} introduces physical roll–yaw interaction through the Γ_4 term, consistent with theoretical predictions. These results confirm the correctness and physical fidelity of the equations of motion implementation.

Part 3: Forces and Moments

A coefficient-based aerodynamic model was developed and integrated with the equations of motion. The model correctly captures lift, drag, side force, and aerodynamic moments as functions of the flow angles, angular rates, and control surface deflections. The thrust was modelled using a linear model appropriate for the Cessna 620 piston engine. Wind disturbances were incorporated by transforming steady wind from the inertial frame to the body frame and adding gust components. Verification through individual control surface deflections confirmed that the model captures both primary control effects and secondary coupling phenomena such as adverse yaw and dihedral effect. The introduction of wind gusts resulted in increased aerodynamic activity, with the behaviour remaining physically consistent and stable throughout.

Part 4: Trim Computation and Linearization

The trim computation was implemented using the Simulink `trim` command as described in Appendix F.1 of [**?**]. The algorithm successfully identified steady-state operating points for three distinct flight conditions:

- **Wings-level flight:** The trim algorithm converged to $\alpha^* = 5.80°$, within the required 5°–7° range. The trim residual $\|\dot{x}^* - f(x^*, u^*)\| \approx 2.63 \times 10^{-13}$ confirms numerical accuracy. All states remained constant in simulation, validating the trim solution.

- **Climb and descent:** The trim algorithm correctly captured steady climb and descent for $\gamma \in \{-0.02, -0.01, 0, 0.02, 0.1\}$ rad. The simulated climb rates matched the theoretical values $\dot{h} = V_a \sin\gamma$ to within numerical precision, confirming that only the altitude changes while all other states remain constant.

- **Coordinated turn:** Trim was computed for load factor $n = 1.2$, yielding a bank angle $\phi^* \approx 33.56°$. The airspeed range corresponding to $C_L \in [0.7, 1.0]$ was derived analytically. Simulation confirmed that the heading ψ increases linearly while all other states remain constant, confirming a steady coordinated turn.

Transfer functions for all eight input-output channels described in Section 5.4 of [**?**] were derived analytically from the trim values. The lateral transfer functions ϕ/δ_a, χ/ϕ, and β/δ_r capture the roll, course, and sideslip dynamics respectively. The longitudinal transfer functions θ/δ_e, h/θ, h/V_a, V_a/δ_t, and V_a/θ capture the pitch, altitude, and airspeed dynamics. The presence of integrators in the roll, pitch, course, and altitude channels confirms the double-integrator structure from control surface to position, which directly motivates the nested loop architecture of the autopilot design in Part 5.

The Simulink model was linearized about the trim condition using the `linmod` command, producing decoupled lateral and longitudinal state-space models. The eigenvalue analysis of A_{lat} and A_{lon} reveals the open-loop modal structure of the aircraft. The consistency check between the analytical transfer functions and those reconstructed from the state-space model confirms the correctness of both approaches.

Overall Assessment

The simulation framework developed in this project successfully captures the nonlinear flight dynamics of the Cessna 620 from first principles. Each component was verified independently. The linearized models derived at trim provide a physically consistent for the autopilot design in Part 5. The systematic approach adopted throughout — moving from coordinate frames to dynamics to forces to trim and linearization — reflects the standard design methodology for unmanned autonomous vehicle guidance and control systems. We regret to leave the fifth part due to time constraints, However, the project was very interesting and fun to do.

6 Team Members Contribution

- **Medhavin (22B0024)** – Completed Section 2 of the assignment, helped in integrating all other tasks and making the report.
- **Amrit (24B0049)** – Completed Section 2 (Kinematics and Dynamics) and contributed to making the report.
- **Mukti (24B0050)** – Completed Section 3 (Forces and Moments) and contributed to making the report.
- **Anay (24B0046)** – Completed Computation and Linearization and contributed to making the report.